

DragonIce, an Initial Full-Body Mixed Reality Experience to Facilitate Interpersonal Synchrony in Children

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ABSTRACT

This study introduces DragonIce, an initial full-body interactive Mixed Reality experience designed, in collaboration with children, to facilitate Interpersonal Synchrony (IPS) among groups of four children. Through an iterative usability study, involving 48 children (8-10 years old), and a mix-method approach to triangulate findings, the research evaluates the system's effectiveness in (i) user-system interaction; (ii) guiding users' movements for IPS; and (iii) overall satisfaction. This iterative design process unveiled insights into synchronization mechanics, informing the development of more sophisticated and enriched interactive experiences. The findings suggest a promising trajectory for assessing the potential impact of IPS on prosocial behavior in the future. Moreover, the study anticipates future exploration of DragonIce's benefits in creating a supportive social environment, particularly for children facing difficulties in social interaction and motor skills, such as those on the Autism Spectrum.

CCS CONCEPTS

- **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Usability Testing;
- **Applied computing** → Law, social and behavioral sciences; Psychology.

KEYWORDS

Full-Body Interaction, Mixed Reality, Interpersonal Synchrony, Autism Spectrum Condition

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1 INTRODUCTION

Imagine a playground where shared laughter, synchronized jumps, and coordinated play gestures not only contribute to playfulness but also establish social bonds among children through a process called Interpersonal Synchrony (IPS) or Interpersonal Entrainment (IPE). IPS, a form of joint action, involves the conscious synchronization between two or more individuals at a common rhythm to perform an activity together [5, 13], such as dancing or marching. This process, functioning as a nonverbal channel of communication and studied mostly without using virtual environments, shows a beneficial impact in early development [3, 4]. Implementing IPS techniques in children has demonstrated increased social bonding [23], helpfulness [3, 12, 24], cooperation [12, 17], and sharing [11, 17, 25] with co-actors, which define prosocial behavior (PB). Therefore, prosocial behavior emerges as a possible catalyst for fostering social interactions among children by cultivating positive social environments.

This phenomenon has been studied mostly in the field of Psychology, with the assistance of an adult in activity facilitation, and recently gaining attention in Human-Computer Interaction (HCI) [7, 14–16, 18–22]. For example, in [18], a music-therapy intervention system utilizing IPS strategies was developed to enhance social and communication skills in children on the Autism Spectrum (AS). Another example, from [21], showcases the potential of a non-interactive full-body system in coordinating the pace and position of four running children, emphasizing how IPS aids attention in neurodevelopmental disorders such as children on the AS. However, in both research fields, there is a research gap in the impact of IPS strategies on groups of more than two individuals. In addition, despite significant advancements in HCI, little attention has been given to understanding how a full-body technology-mediated environment can be used to facilitate IPS and effectively foster PB in groups of more than two children and without the assistance of an adult. Technologies employing full-body interactions emerge as highly suitable for their promotion because of their benefits

of promoting intuitive, dynamic, natural HCIs, and face-to-face social interaction between multiple users. Mixed Reality (MR) technologies, in particular, make it possible to create applications in which we interact with the whole body. Consequently, users can relate their actions in space to the movements of others within a full-body scale environment creating an ecologically valid environment that has analogous properties to a school playground, with the advantage of being able to provide a broad range of dynamic environments to engage all children. These interactions are based on the theories of Embodied Cognition [6, 10, 26], emphasizing the formative role of the body and the environment in cognitive processes. Exploring and integrating IPS techniques, particularly in the context of full-body MR environments, holds promise for advancing our understanding of prosocial behavior to foster inclusive social interactions among groups of children.

2 OBJECTIVE AND RESEARCH QUESTIONS

The present study aimed to develop an initial interactive full-body MR experience, DragonIce, to mediate IPS among four children between 8-10 years old, without requiring an external human facilitator. To evaluate its effectiveness, we conducted an iterative usability analysis over two sessions, focusing on addressing three primary research questions (RQ): (RQ1) How does each child interact with the system? (individual performance); (RQ2) How does the system guide four users to synchronize their movements? (group performance); and (RQ3) How satisfied are users after playing the interactive experience? Additionally, after performing the first usability study, we identified features to be improved by not obtaining the expected results for RQ1 and RQ2. We modified it and conducted a new usability study to investigate: (RQ4) Do the changes improve the results obtained in RQ1 and RQ2? and (RQ5) Do the changes maintain or improve user satisfaction (RQ3)? By rigorously exploring these RQs, we seek to contribute valuable insights into the intersection of HCI, IPS, and child development, fostering a richer understanding of how technology can be harnessed to foster social interaction.

3 METHODS

3.1 Study, Demographics and Ethics

A between-subjects design was employed to test the initial (first iteration) and the refined (second iteration) prototypes. We conducted the trials with 50 third-grade students (8-10 years old) distributed in two regular public school groups, who had previously participated in the co-design of DragonIce [8]. The usability sessions took place with each school group at one-week intervals, accommodating 25 children per session. The inclusion criteria were that the children could understand and follow the instructions well and be in a group of four. The tutor, as the person who knows the children best, formed the subgroups, ensuring no strong relationships between them (positive or negative). Sex and gender were not considered in the grouping. The children in the groups knew each other to create a situation analogous to that of traditional small-scale societies [12]. Two children were excluded because their group consisted of peers who had already played the experience, resulting in a final group size of 48 children (19 females (sex assigned at birth); 18 girls (gender as a social role); 26 males/boys (sex/gender); 2 bi-gender

(gender); 3 and 2 prefer not to answer (sex and gender); Mage = 8 years old, SDage = 0.62 years old). The research was conducted under the approval of Pompeu Fabra University's ethics board. We gathered written informed consent forms from the legal representatives before the sessions and oral assent from the children on the day of the session.

3.2 General Experimental Design

DragonIce was developed through a systematic process: (1) defining the initial design requirements; (2) conducting co-design sessions with children where they designed the narrative, characters, visual elements, and actions of the video game; and (3) carrying out an iterative evaluation approach for the initial DragonIce prototype resulting from the co-design sessions. This study examines the third step, the iterative evaluation process. For comprehensive details on the design requirements, the co-design sessions, and their evaluation refer to [8].

3.2.1 Iterative Evaluation Approach: Usability Studies. After conducting the co-design sessions, the main objective was to assess through an iterative design approach the effectiveness of the system in facilitating synchronized movements among four users. The initial prototype was refined based on direct observations, user perceptions, and user/group performance. Each session consisted of three stages, lasting approximately 30 minutes for each subgroup: (1) a pre-questionnaire (10mins); (2) the performance of the joint task in the initial DragonIce prototype (10mins); and (3) a post-questionnaire (10mins). During the waiting periods, the subgroups engaged in activities such as creating souvenir badges, reflective drawings, and brainstorming possible second levels for the interactive experience. In stage 1, children completed a short questionnaire assessing group-cohesion, perceived relationships, demographic data, and dancing and music abilities (see Supplementary Material Table-1 for details). Afterward, they were taken to a nearby room to carry out the initial DragonIce prototype (stage 2). Immediately after, in an adjacent room, the children completed a post-questionnaire during stage 3. This questionnaire included sections on skills and affiliation, user-experience (UX), and design outcomes (see Supplementary Material Table-1 for details). Figure 1 shows the different stages carried out during the session.

Full-body Mixed Reality Experience: Initial DragonIce Prototype. The performance of the joint task, stage 2, was performed in a 6x6 meters floor projection MR system designed to provide face-to-face non-invasive experiences (Figure 2). The MR system comprises a graphic workstation, two HD projectors generating a 1920x1920 pixels floor projection, a HiFi spatially-panned sound system, and our in-house computer vision-based tracking system using "light markers" (Figure 2). These interactive objects are battery-powered, luminous pointers, secured to each user's right foot (Figure 2) with four distinct LED colors: red, green, blue, and magenta. The application of the experience was developed in Unity and C#. The shared goal was to liberate a dragon that was trapped under the ice of a frozen lake, consisting of three phases: (1) familiarization with the environment and co-actors; (2) joint task of synchronized movement; and (3) experience closure (see Figure 1).

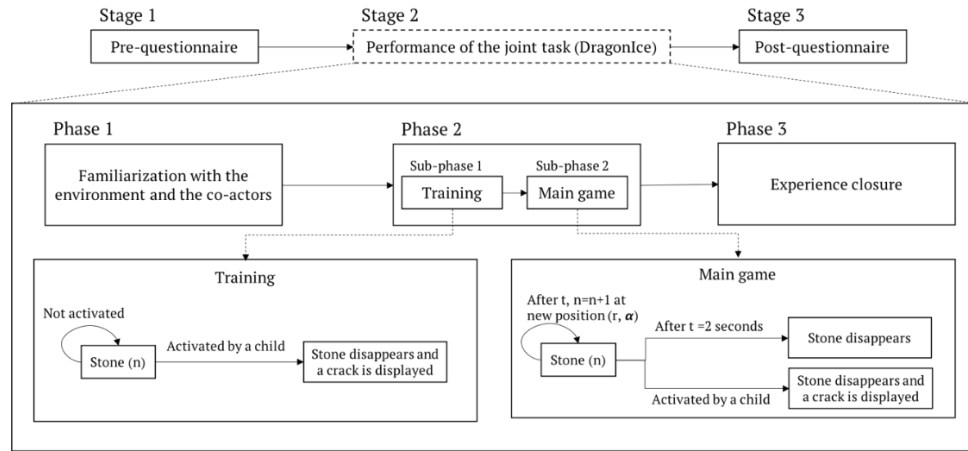


Figure 1: Diagram with an overview of the stages that were carried out during the session by each group of four children and a more detailed view of the phases during the joint interactive experience (stage 2).

The experience starts by showing a central lake with four wooden docks on the four corners. The children stand on designated docks. They then witness how two evil characters freeze the lake trapping the dragon under the ice surface. The position of the children allows them to watch the totality of the virtual environment and their peers, fostering familiarity before they start playing; this will be essential in the future for children on the Autism Spectrum. The system detects dock occupancy to trigger the next phase. The joint activity (phase 2) has two sub-phases. The first sub-phase provides a short training sequence to familiarize children with the game mechanics using the light marker on their foot. The dragon, located at the center of the lake beneath the ice, throws magic stones matching the color of each player's light marker. Each stone glides under the ice from the center of the lake to a spot in front of each user's dock. Once the stone stops, the children must then step on their stone with the foot holding their light marker. This creates a crack in the ice over the stone producing a cracking sound and a visual effect. The stone disappears the moment the child steps on it. Once all four children perform successfully the action, the dragon then throws four new stones in a new trajectory which is traced at an angle from the first in an anti-clockwise sense (to the right of each user). Users learn that stones will keep appearing always slightly to the right of their last position. After this training sub-phase, the second one is the main game. The dragon now throws stones at defined intervals but, unlike the previous phase, the stone can be stepped on only within a time slot of two seconds before it disappears, encouraging a dynamic movement. A total of 60 stones were thrown during the 5-minute duration of the game. Moreover, the separation angle between two consecutive final positions was always 30° (Figure 3). Our iterative design process defined two different strategies for the final position of the stones: "random final position" (first iteration) and "spiral final position" (second iteration). In the "random final position", every new stone ends its trajectory at a random distance from the center, between a minimum and a maximum radius ($r = [18, 35]$ units). Moreover, the constant interval time between stones was defined as $t=5$ seconds. However, due to the results obtained during the

first usability test, the mechanics were modified to a "spiral final position". The refined mechanics start with a maximum radius ($r_{\max} = 35$ units) for the first stone's trajectory that gradually decreases to a minimum radius with every new stone ($r_{\min} = 18$ units). In this strategy, the children move progressively closer to the center, and are increasingly capable of perceiving the position and movement of their peers, which is important for acquiring an IPS social outcome. In this new iteration, the interval time for throwing a new stone also progressively decreases from $t=6$ seconds to $t=3$ seconds to make the experience progressively faster and hence provide a more intense tempo for the users, which is meant to benefit the sense of synchronicity in the users. Figure 3 shows an example graph of each mechanic. Two more game aspects were modified in the improved prototype. The initial version featured a square object indicating each user's position and two stones in the training sub-phase, while the second version omitted the square object and provided only one training stone. After all 60 stones are thrown in the main game subphase, phase 3 begins, requiring children to return to their docks to see the end of the story. They observe how the cracks completely spread throughout the surface of the ice, which, thanks to the dragon's final tail strike, completely breaks allowing the dragon to escape. This ending was designed to prevent frustration, ensuring that all children, regardless of their performance, achieve the final objective of freeing the dragon. The visuals used enable a dynamic playful experience, where the movements of the four players are guided by the system without direct external mediation by an adult. Additional details on the dimensions of IPS used in the initial prototype are provided in Supplementary Material Table-2.

Data Gathering. Data was gathered through observations, system-generated data, and surveys utilizing a mixed-method approach to triangulate findings and achieve a holistic view of the system's effectiveness in synchronizing users' movements and satisfaction. Two of the project's researchers documented observation data for each subgroup, analyzing children's task performance (user-system interaction, movement accuracy, and interactive object construction), attention focus, and player engagement. It provided

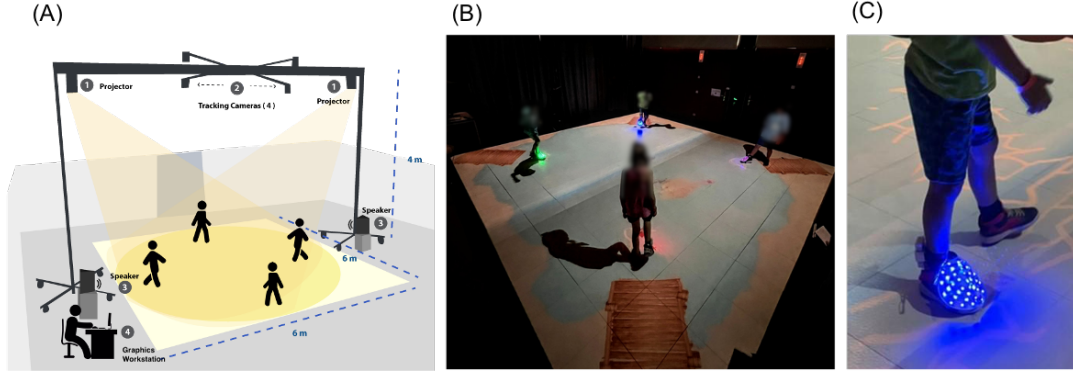


Figure 2: (A) Schema of the physical configuration of the system. (B) Initial Dragonice prototype with four children performing the activity. (C) Luminous interactive object placed on the child's foot and used to interact with the virtual environment.

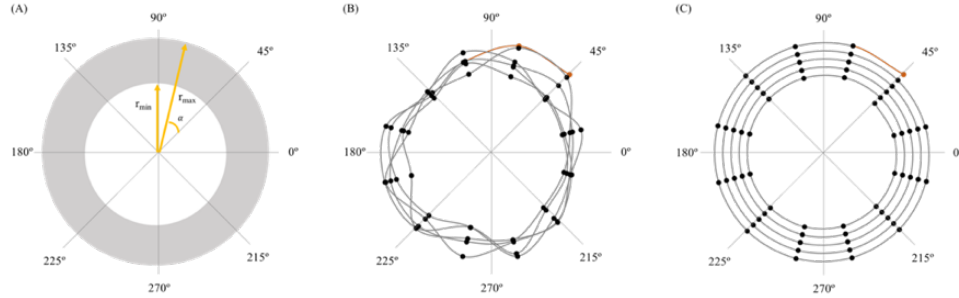


Figure 3: (A) Graph showing the space, gray zone, where a stone can be placed. The r_{min} indicates the minimum distance between the final position of a stone and the center, with a value of 18 units, and r_{max} the maximum distance between the final position of a stone and the center, with a value of 35 units. Alpha (α) indicates the angular separation between consecutive stones, with a value of 30°. (B) Graph of the expected movement during the first iteration made by a child following and stepping on the stones. The final position of the stone follows the "random final position" strategy where the distance between the stone's final position and the center is a random value between 18 and 35 units. (C) Graph of the expected movement during the second iteration made by a child following and stepping on the stones. The final position of the stone follows the "spiral final position" strategy where the distance between the stone's final position and the center decreases progressively from 35 units to 18 units. In graphs B and C, the points indicate the stone's final position, the orange color represents the training sub-phase, and the black color represents the main game sub-phase.

qualitative insights into user-system and user-user interaction that are important for achieving IPS [19]. The system-generated data recorded whether or not users activated the stones, which provided objective quantitative data on user-system interaction. In addition, a UX questionnaire on a laptop with a two-sided slider ("Not at all" (0) - "A lot" (100)) obtained quantitative information on ease of use, satisfaction, performance, and visual design. The answers from the questionnaires were treated as continuous data.

4 RESULTS

For comparisons between iterations, RQ4 and RQ5, the data distribution normality was assessed using the Shapiro-Wilk test. For normally distributed data, an independent sample t-test was conducted ($p < 0.05$). Non-normally distributed data underwent the

Mann-Whitney U test. JASP (Jeffrey's Amazing Statistics Program) software was used for the data analysis.

4.1 User-system Interaction

In addressing the RQ1 on individual user-system interaction, we assessed how effectively each child could interact with the system. We examined the frequency of stone activations by each user, made observations on their interactions with the system, and gathered feedback on the ease of play, understanding when to move, and knowing what to do. In the first iteration, the mean (M) and standard deviation (SD) of stone activations frequencies were 0.268 and 0.121, respectively, while the second iteration showed higher values with $M = 0.611$ and $SD = 0.165$. Detailed results for user-system interaction observations are provided in Table 1. Children's perceptions in the first iteration included $Mdn = 83$, $M = 73.417$ and SD

Table 1: General findings of the observed interactions of the children with the MR system.

Observation category	Observation sub-category	Findings of the first usability study	Findings of the second usability study
Performance	User-system Interaction	Children followed the experience well, although behaviors such as stepping on the stone more than once and very hard were observed. Also, some users stepped on the visual symbol of the tracking system instead of the stone.	Children interacted with the system well. It was noted that some children did not want to leave the dock at the start and they were stepping on the stones very hard.
	Movement accuracy	Players stepped on the stone with anticipation, especially when the final position of the new stone (n+1) was more distant from the center than the previous stone (n).	Not much anticipation of stepping on the stone was observed and when it occurred, it was at the beginning of the activity.
	Interactive Object Constriction	All groups had problems with the LED lights of the interactive objects, momentarily stopping the dynamics created during the experience.	There were some issues with the interactive objects, but in general, they did not stop the game dynamics created.
Attention Focus		Children primarily focused on the thrown stone and occasionally turned their backs, especially when the stone was directed further backward from its previous position.	Children looked more at the center and their peers. It was also observed that they were focusing their gaze on the stone.
Engagement		Children made expressions such as "Wow" when the dragon came out, and they were touching the dragon at the end or jumping into the water to pretend to swim.	Expressions such as "Look at the lake", once the cracks were created, and "It is so cool, I love it". Also, they were touching the dragon at the end of the experience.

= 30.006 for ease of play, $Mdn = 97.5$, $M = 84.909$ and $SD = 23.322$ for understanding when to move, $Mdn = 98.5$, $M = 80.625$ and $SD = 30.229$ for knowing what to do. In the second iteration, these scores were $Mdn = 80.5$, $M = 76.417$ and $SD = 26.655$, $Mdn = 99$, $M = 92.542$ and $SD = 14.090$, $Mdn = 100$, $M = 93.478$ and $SD = 15.974$, respectively. All means were above a value of 70 and close to the maximum value of 100, being positively skewed.

Scores on stone activations frequencies in the first iteration and ease of play, understanding when to move, and knowing what to do in both iterations were not normally distributed, as assessed by Shapiro-Wilk's test ($p < 0.05$). A Mann-Whitney U test was run to determine if there were differences between the scores of both iterations (RQ4). Stone activation frequencies score was statistically significantly different between first iteration ($Mdn = 0.230$) and second iteration ($Mdn = 0.665$), $U = 37$, $p < 0.001$, Cohen's $d = -0.872$). In contrast, none of the user perceptions (ease of play, understanding when to move, and knowing what to do) showed statistically significant differences between the two iterations.

4.2 System Facilitating Interpersonal Synchrony

In addressing the RQ2, which explores the system's efficacy in guiding users to synchronize their movements, we analyzed the frequency of stones activated simultaneously by three or four players (group performance); the observations made on movement accuracy, interactive object constriction, and attention focus; and user perceptions on whether they noticed their peers while moving and whether they perceived synchronization in their movements. The group performance data were normalized to a ratio between 0 (minimum) and 1 (maximum). The first iteration showed a group interaction with a mean (M) of 0.128 and a standard deviation (SD)

of 0.118, while the second iteration exhibited $M = 0.497$ and $SD = 0.185$. The corresponding observation results are detailed in Table 1. Concerning children's perceptions, in the first iteration, scores were $Mdn = 96$, $M = 80.478$ and $SD = 26.685$ for peer noticing and $Mdn = 65$, $M = 56.609$ and $SD = 34.840$ for perceiving simultaneous movement. In the second iteration, these scores were $Mdn = 97.5$, $M = 82.958$ and $SD = 27.847$ and $Mdn = 58.5$, $M = 65.958$ and $SD = 28.816$, respectively.

Scores on group performance in both iterations were normally distributed, as assessed by Shapiro-Wilk's test ($p > 0.05$). An independent sample t-test, comparing stone activation frequencies by three or four players simultaneously between the two iterations (RQ4), revealed that the results in the second iteration ($M = 0.497$, $SD = 0.185$) were statistically significantly higher than in the first one ($M = 0.128$, $SD = 0.118$), $t(10) = -4.130$, $p = 0.002$, Cohen's $d = -2.392$). Scores on user perceptions in both iterations were not normally distributed, as assessed by Shapiro-Wilk's test ($p < 0.05$). A Mann-Whitney U test revealed no statistically significant differences in none of the user perceptions (noticing peers and perceiving synchronization) between the two iterations.

4.3 User Satisfaction

In addressing the RQ3, which explores user satisfaction with the interactive experience, we analyzed the users' answers on whether they enjoyed playing the game and whether they would play the game again. In the first iteration, children's perceptions yielded scores of $Mdn = 100$, $M = 92.375$ and $SD = 15.937$ for enjoyment and $Mdn = 100$, $M = 92.125$ and $SD = 16.794$ for willingness to play again. In the second iteration, these scores were $Mdn = 100$, $M = 95.478$ and $SD = 18.161$ for enjoyment and $Mdn = 100$, $M = 96.375$

and $SD = 15.550$ for willingness to play again. All means were above a value of 90, being very close to the maximum value of 100.

Scores on user perceptions in both iterations were not normally distributed, as assessed by Shapiro-Wilk's test ($p < 0.05$). A Mann-Whitney U test revealed no statistically significant differences in none of the user perceptions (enjoying playing the game and wanting to play the game again) between the two iterations.

5 DISCUSSION

The main goal of this study was to present initial findings on the development of Mixed Reality (MR) full-body experiences designed to facilitate Interpersonal Synchrony (IPS) among four children, eliminating the need for an external human facilitator. To achieve this objective, we conducted an iterative usability study comprising two iterations to evaluate the effectiveness of DragonIce, a playful full-body MR interactive experience aiming to facilitate IPS.

The iterative evaluation of the system's ability to promote synchronization involved a comprehensive analysis of the user-system interaction, including individual stone activations (individual performance), direct observations, and user feedback across two usability sessions. The system-generated data showed that during the second iteration, the enhanced mechanics significantly improved user performance, with a 34% increase in stone activations. The results suggested positive effects of the refined game mechanics on user-system interaction. Observations from the first iteration revealed that children generally followed the experience, but some displayed behaviors like excessive force and stepping on the stone multiple times. Confusion also arose, with some users mistakenly stepping on the tracking system symbol. In the second iteration, while the children interacted well, new observations emerged. Some hesitated to leave the dock initially, and instances of excessive force while stepping on stones were noted. This iterative approach allows us to address existing challenges while revealing new aspects to consider. Additionally, user feedback provided valuable qualitative insights into user-system interaction, indicating positive trends in ease of play, knowing when to move, and understanding of what to do across both iterations. The positively skewed distributions support the overall satisfaction and comfort of the participants with the system. The results, both quantitative and qualitative, indicated an overall positive trend in user experience. The observations underscore the importance of continuous refinement to enhance user understanding. However, while stone activation frequencies significantly improved, subjective evaluations remained relatively consistent between iterations.

As the purpose of DragonIce is to guide users to synchronize their movements, we analyzed the group performance based on the system-generated data, the direct observations, and the user's feedback. In line with the user performance findings, the interactive experience demonstrated enhanced group performance during the second iteration. A significant improvement of 37% was noted, indicating successful coordination 50% of the time. Design refinements, such as the progressive reduction of interval time and final position patterns, contributed to participants paying closer attention to peers' movements and avoiding undesired movements, which are important for acquiring the effect of interpersonal synchrony [19].

The results derived from the UX questionnaire indicate that challenges in sustaining coordination during the game were not linked to comprehension or awareness but rather users perceived that they were moving at a low level of synchrony, as assessed in the group performance evaluation. Based on the observations, the complexities of sustaining synchrony can be attributed to different factors, either individually or in combination. The first factor explores the influence of a player on others, known as the perception-behavior link [2]. Hence, when a player encounters difficulty in executing an activity adequately, their performance may influence that of the other players. The second factor centers on the tempo employed for synchronization. In comparison to previous research (for a review see [5]), our study employed a slower tempo, which might lead to less control over movement, unintended or additional superfluous interaction behaviors, and/or less perceived entrainment by not indicating the intention of the synchronization at the start. Notably, undesired behaviors such as movement anticipations during stone activation were observed, which affect the movement synchrony between the four players as they do not all perform the same movement at the same time. Despite challenges in guiding user's movements perfectly, high user satisfaction levels were reported in both iterations. Participants expressed enjoyment and interest in playing the game again, highlighting the engaging nature of the DragonIce experience.

5.1 Limitations and Future Work

This study provides valuable findings on the intersection between the fields of HCI, IPS, and child development, although some limitations need to be considered. Limitations included issues with the interactive object's robustness during the first iteration and not considering sex and/or gender in the grouping to analyze their interactive differences. Future research should explore alternative interactive objects and consider sex and/or gender in individual and group performance. These limitations present promising avenues for further research on IPS techniques to foster and maintain synchronization.

Additionally, there is a promising trajectory to explore the potential benefits of facilitating IPS, particularly in creating a more conducive social environment for children facing difficulties in social interaction and motor skills and with an affinity for Information and Communication Technologies (ICT), such as those on the Autism Spectrum [1, 9].

6 CONCLUSION

This study presents the preliminary outcomes of DragonIce, designed to facilitate IPS by employing an engaging full-body interactive MR experience to guide the movements of a group of four children. The iterative design process employed allowed us to move towards a better understanding of the mechanics that enhance synchronization among users. These findings guide us in advancing towards the development of enhanced interactive experiences that facilitate IPS, with a focus on assessing their potential impact on PB in the future. This study lays the groundwork for future research exploring technology's transformative role in fostering positive social contexts for diverse groups of children.

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REFERENCES

- [1] American Psychiatric Association. 2013. *American Psychiatric Association, 2013. Diagnostic and statistical manual of mental disorders (5th ed.)*. DOI:<https://doi.org/10.1176/appi.books.9780890425596.744053>
- [2] Tanya L. Chartrand and John A. Bargh. 1999. The Chameleon Effect: The Perception-Behavior Link and Social Interaction. *Personal. Soc. Psychol.* 76, 6 (1999), 893–910.
- [3] Laura K. Cirelli. 2018. How interpersonal synchrony facilitates early prosocial behavior. *Curr. Opin. Psychol.* 20, (April 2018), 35–39. DOI:<https://doi.org/10.1016/J.COPSYC.2017.08.009>
- [4] Laura K. Cirelli, Stephanie J. Wan, and Laurel J. Trainor. 2014. Fourteen-month-old infants use interpersonal synchrony as a cue to direct helpfulness. *Philos. Trans. R. Soc. B Biol. Sci.* 369, 1658 (December 2014). DOI:<https://doi.org/10.1098/RSTB.2013.0400>
- [5] Liam Cross, Martine Turgeon, and Gray Atherton. 2019. How Moving Together Binds Us Together: The Social Consequences of Interpersonal Entrainment and Group Processes. *Open Psychol.* 1, 1 (2019).
- [6] Paul Dourish. 2001. Where the Action Is: The Foundations of Embodied Interaction. *Where action is Found. embodied Interact.* (2001). DOI:<https://doi.org/10.1162/leon.2003.36.5.412>
- [7] Olga Gali, Sena Beste Ercan, Gray Atherton, Liam Cross, Batuhan Sayis, Pamela Heaton, and Narcis Pares. 2023. Mediating Interpersonal Synchronization in Children through a Full-Body Mixed Reality System: Analysis of the Pre-Interactive Mandala Experience. *PRESENCE Virtual Augment. Real.* (July 2023), 1–17. DOI:https://doi.org/10.1162/pres_a_00386
- [8] Olga Gali, Sena Beste Ercan, Deniz Hepdoğan, Gray Atherton, Liam Cross, and Narcis Pares. 2023. *Inclusive Embodied Participatory Design Procedure with and for Children with Diverse Functional Abilities to Foster Interpersonal Synchrony in a Mixed Reality Experience*. Manuscript submitted for publication.
- [9] Tina R. Goldsmith and Linda A. LeBlanc. 2004. Use of technology in interventions for children with autism. *J. Early Intensive Behav. Interv.* 1, 2 (2004), 166–178. DOI:<https://doi.org/10.1037/h0100287>
- [10] Joseph Grady. 2002. Philosophy in the Flesh: The Embodied Mind and its Challenge to Western Thought. *J. Pragmat.* 34, 12 (2002), 1903–1909. DOI:[https://doi.org/10.1016/S0378-2166\(02\)00058-9](https://doi.org/10.1016/S0378-2166(02)00058-9)
- [11] Sebastian Kirschner and Beatriz Ilari. 2014. Joint Drumming in Brazilian and German Preschool Children. *J. Cross. Cult. Psychol.* 45, 1 (January 2014), 137–166. DOI:<https://doi.org/10.1177/0022022113493139>
- [12] Sebastian Kirschner and Michael Tomasello. 2010. Joint music making promotes prosocial behavior in 4-year-old children. *Evol. Hum. Behav.* 31, 5 (September 2010), 354–364. DOI:<https://doi.org/10.1016/j.evolhumbehav.2010.04.004>
- [13] Leonie Koban, Anand Ramamoorthy, and Ivana Konvalinka. 2019. Why do we fall into sync with others? Interpersonal synchronization and the brain's optimization principle. *Soc. Neurosci.* 14, 1 (2019), 1–9. DOI:<https://doi.org/10.1080/17470919.2017.1400463>
- [14] Tamara Lorenz, Alexander Mörtl, and Sandra Hirche. 2013. Movement synchronization fails during non-adaptive human-robot interaction. *ACM/IEEE Int. Conf. Human-Robot Interact.* (2013), 189–190. DOI:<https://doi.org/10.1109/HRI.2013.6483565>
- [15] Patrick Nalepka, Rachel W. Kallen, Anthony Chemero, Elliot Saltzman, and Michael J. Richardson. 2017. Herd Those Sheep: Emergent Multiagent Coordination and Behavioral-Mode Switching. *Psychol. Sci.* 28, 5 (2017), 630–650. DOI:<https://doi.org/10.1177/0956797617692107>
- [16] Patrick Nalepka, Maurice Lamb, Rachel W. Kallen, Kevin Shockley, Anthony Chemero, Elliot Saltzman, and Michael J. Richardson. 2019. Human social motor solutions for human-machine interaction in dynamical task contexts. *Proc. Natl. Acad. Sci.* 116, 4 (January 2019), 1437–1446. DOI:<https://doi.org/10.1073/PNAS.1813164116>
- [17] Tal-Chen Rabinowitch and Andrew N. Meltzoff. 2017. Joint Rhythmic Movement Increases 4-Year-Old Children's Prosocial Sharing and Fairness Toward Peers. *Front. Psychol.* 0, JUN (June 2017), 1050. DOI:<https://doi.org/10.3389/FPSYG.2017.01050>
- [18] Grazia Ragone, Kate Howland, and Emeline Brulé. 2022. Evaluating Interactional Synchrony in Full-Body Interaction with Autistic Children. *Proc. Interact. Des. Child. IDC 2022* (2022), 1–12. DOI:<https://doi.org/10.1145/3501712.3529729>
- [19] Michal Rinott and Noam Tractinsky. 2021. Designing for interpersonal motor synchronization. *Human-Computer Interact.* (2021). DOI:<https://doi.org/10.1080/07370024.2021.1912608>
- [20] Ekaterina R. Stepanova, John Desnoyers-Stewart, Philippe Pasquier, and Bernhard E. Riecke. 2020. JeL: Breathing together to connect with others and nature. *DIS 2020 - Proc. 2020 ACM Des. Interact. Syst. Conf.* (July 2020), 641–654. DOI:<https://doi.org/10.1145/3357236.3395532>
- [21] Issey Takahashi, Mika Oki, Baptiste Bourreau, Itaru Kitahara, and Kenji Suzuki. 2018. FUTUREGYM: A gymnasium with interactive floor projection for children with special needs. *Int. J. Child-Computer Interact.* 15, (2018), 37–47. DOI:<https://doi.org/10.1016/j.ijcci.2017.12.002>
- [22] Bronwyn Tarr, M. Slater, and Emma Cohen. 2018. Synchrony and social connection in immersive Virtual Reality. *Sci. Rep.* 8, 1 (2018), 1–9. DOI:<https://doi.org/10.1038/s41598-018-21765-4>
- [23] Bahar Tunçgenç and Emma Cohen. 2016. Movement Synchrony Forges Social Bonds across Group Divides. *Front. Psychol.* 7, (May 2016). DOI:<https://doi.org/10.3389/fpsyg.2016.00782>
- [24] Bahar Tunçgenç and Emma Cohen. 2018. Interpersonal movement synchrony facilitates pro-social behavior in children's peer-play. *Dev. Sci.* 21, 1 (January 2018), e12505. DOI:<https://doi.org/10.1111/DESC.12505>
- [25] Yingjia Wan and Liqi Zhu. 2021. Effects of rhythmic turn-taking coordination on five-year-old children's prosocial behaviors. *Dev. Psychol.* 57, 11 (November 2021), 1787–1795. DOI:<https://doi.org/10.1037/dev0001244>
- [26] Margaret Wilson. 2002. Six views of embodied cognition. *Psychonomic Bulletin and Review* 9, 625–636. DOI:<https://doi.org/10.3758/BF03196322>